

Assignment

1) What are the types of Applications?

Ans:

Common types of applications are:

- Desktop Applications – MS Word, Calculator
- Web Applications – Gmail, Facebook
- Mobile Applications – WhatsApp, Instagram
- Enterprise Applications – ERP, Banking software
- Embedded Applications – Washing machine software

2) What is programing?

Ans:

Programming is the process of **writing instructions (code)** for a computer to perform a specific task using a programming language.

3) What is Python?

Ans:

Python is a high-level, interpreted, object-oriented programming language that is:

- Easy to learn
- Used in web development, data science, AI, automation, etc.

4) Write a Python program to check if a number is positive, negative or zero.

Ans:

```
] num = int(input("Enter a number: "))

if num > 0:
    print("Positive number")
elif num < 0:
    print("Negative number")
else:
    print("Zero")
```

Enter a number: 7
Positive number

5)Write a Python program to get the Factorial number of given numbers.

Ans:

```
: num = int(input("Enter a number: "))
fact = 1

for i in range(1, num + 1):
    fact *= i

print("Factorial is:", fact)
```

Enter a number: 4
Factorial is: 24

6)Write a Python program to get the Fibonacci series of given range.

Ans:

```
n = int(input("Enter range: "))
a, b = 0, 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

Enter range: 9
0 1 1 2 3 5 8 13 21

7)How memory is managed in Python?

Ans:

- Python uses **automatic memory management**
- Uses **Heap memory**
- Uses **Garbage Collector** to delete unused objects

8)What is the purpose continuing statement in python?

Ans:

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

1
2
4
5

9) Write python program that swap two number with temp variable and without temp variable.

Ans:

```
a = 10
b = 20
temp = a
a = b
b = temp
print(a, b)
```

20 10

Without temp variable

```
a = 10
b = 20
a, b = b, a
print(a, b)
```

20 10

10) Write a Python program to find whether a given number is even or odd, print out an appropriate message to the user.

Ans:

```
num = int(input("Enter number: "))

if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

Enter number: 18
Even number

11) Write a Python program to test whether a passed letter is a vowel or not.

Ans:

```
ch = input("Enter a letter: ")

if ch.lower() in "aeiou":
    print("Vowel")
else:
    print("Not a vowel")
```

Enter a letter: 6
Not a vowel

12) Write a Python program to sum of three given integers. However, if two values are equal sum will be zero.

Ans:

```
: a = int(input())
  b = int(input())
  c = int(input())

if a == b or b == c or a == c:
    print(0)
else:
    print(a + b + c)
```

89
90
76
255

13) Write a Python program that will return true if the two given integer values are equal or their sum or difference is 5.

Ans:

```
[1]: a = int(input())
      b = int(input())

      if a == b or a + b == 5 or abs(a - b) == 5:
          print(True)
      else:
          print(False)
```

```
67
90
False
```

```
[ ]:
```

14)Write a python program to sum of the first n positive integers.

Ans:

```
!]: n = int(input("Enter n: "))
      total = n * (n + 1) // 2
      print("Sum is:", total)
```

```
Enter n: 78
Sum is: 3081
```

15)Write a Python program to calculate the length of a string.

Ans:

```
s = input("Enter string: ")
print("Length:", len(s))
```

```
Enter string: 888
Length: 3
```

16) Write a Python program to count the number of characters (character frequency) in a string.

Ans:

```
: s = input("Enter string: ")

for char in s:
    print(char, ":", s.count(char))
```

Enter string: 780

7 : 1

8 : 1

0 : 1

17)What are negative indexes and why are they used?

Ans:

Negative indexes access elements from the end of a sequence.

- -1 → last element
- -2 → second last

Used to easily access elements without knowing length.

18)Write a Python program to count occurrences of a substring in a string.

Ans:

```
: s = input("Enter string: ")
  sub = input("Enter substring: ")

print(s.count(sub))
```

Enter string: drashti

Enter substring: soni

0

19) Write a Python program to count the occurrences of each word in a.

Ans:

```
s = input("Enter sentence: ")
words = s.split()

for word in words:
    print(word, ":", words.count(word))
```

```
Enter sentence: soni
soni : 1
```

20) Write a Python program to get a single string from two given strings, separated by a space and swap the first two characters of each string.

Ans:

```
: s1 = input("Enter first string: ")
   s2 = input("Enter second string: ")

   new_s1 = s2[:2] + s1[2:]
   new_s2 = s1[:2] + s2[2:]

   print(new_s1 + " " + new_s2)
```

```
Enter first string: diya
Enter second string: drashti
drya diashti
```


21) Write a Python program to add 'in' at the end of a given string (length should be at least 3). If the given string already ends with 'ing' then add 'ly' instead if the string length of the given string is less than 3, leave it unchanged.

Ans:

```
: s = "play"

if len(s) < 3:
    print(s)
elif s.endswith("ing"):
    print(s + "ly")
else:
    print(s + "in")
```

playin

22) Write a Python function to reverse a string if its length is a multiple of 4.

Ans:

```
: s = "abcd"

if len(s) % 4 == 0:
    print(s[::-1])
else:
    print(s)
```

dcba

: |

23) Write a Python program to get a string made of the first 2 and the last 2 chars from a given a string. If the string length is less than 2, return instead of the empty string.

Ans:

```
: s = "python"

if len(s) < 2:
    print("")
else:
    print(s[:2] + s[-2:])
```

pyon

24)Write a Python function to insert a string in the middle of a string.

Ans:

```
s1 = "<<>>"
s2 = "Python"

mid = len(s1) // 2
print(s1[:mid] + s2 + s1[mid:])
```

<<Python>>

25)What is List? How will you reverse a list?

Ans:

```
lst = [1, 2, 3]
lst.reverse()
print(lst)
```

[3, 2, 1]

26)How will you remove last object from a list?

Ans:

```
: lst = [10, 20, 30]
  lst.pop()
  print(lst)
```

[10, 20]

27)Suppose list1 is [2, 33, 222, 14, and 25], what is list1 [- 1]?

Ans:

25

```
list1 = [2, 33, 222, 14, 25]
print(list1[-1])
```

25

28)Differentiate between append () and extend () methods?

Ans:

```
a = [1, 2]
a.append(3)
print(a)

b = [1, 2]
b.extend([3, 4])
print(b)
```

```
[1, 2, 3]
[1, 2, 3, 4]
```

29) Write a Python function to get the largest number, smallest num and sum of all from a list.

Ans:

```
: lst = [10, 20, 5, 40]

print("Largest:", max(lst))
print("Smallest:", min(lst))
print("Sum:", sum(lst))
```

```
Largest: 40
Smallest: 5
Sum: 75
```

30)How will you compare two lists?

Ans:

```
: l1 = [1, 2, 3]
  l2 = [1, 2, 3]

print(l1 == l2)
```

```
True
```

31) Write a Python program to count the number of strings where the string length is 2 or more and the first and last character are same from a given list of strings.

Ans:

```
: lst = ["abc", "aba", "121", "xyz"]
count = 0

for s in lst:
    if len(s) >= 2 and s[0] == s[-1]:
        count += 1

print(count)
```

2

32)Write a Python program to remove duplicates from a list.

Ans:

```
lst = [1, 2, 2, 3, 3]
print(list(set(lst)))
```

[1, 2, 3]

|

33)Write a Python program to check a list is empty or not.

Ans:

```
lst = []  
  
if len(lst) == 0:  
    print("List is empty")
```

List is empty

34) Write a Python function that takes two lists and returns true if they have at least one common member.

Ans:

```
l1 = [1, 2, 3]  
l2 = [3, 4, 5]  
  
for i in l1:  
    if i in l2:  
        print(True)  
        break
```

True

35) Write a Python program to generate and print a list of first and last 5 elements where the values are square of numbers between 1 and 30.

Ans:

```
squares = []

for i in range(1, 31):
    squares.append(i*i)

print(squares[:5])
print(squares[-5:])
```

```
[1, 4, 9, 16, 25]
[676, 729, 784, 841, 900]
```

36) Write a Python function that takes a list and returns a new list with unique elements of the first list.

Ans:

```
lst = [1, 2, 2, 3]
print(list(set(lst)))
```

```
[1, 2, 3]
```

|

37) Write a Python program to convert a list of characters into a string.

Ans:

```
lst = ['P', 'y', 't', 'h', 'o', 'n']
s = ""

for i in lst:
    s += i

print(s)
```

Python

38)Write a Python program to select an item randomly from a list.

Ans:

```
: import random

lst = [10, 20, 30, 40]
print(random.choice(lst))
```

20

39)Write a Python program to find the second smallest number in a list.

Ans:

```
: lst = [10, 5, 20, 8]
lst.sort()
print(lst[1])
```

8

40)Write a Python program to get unique values from a list

Ans:


```
lst = [1, 2, 2, 3, 4]
print(list(set(lst)))
```

[1, 2, 3, 4]

41)Write a Python program to check whether a list contains a sub list

Ans:

```
] : def is_sublist(main_list, sub_list):
    n = len(sub_list)
    for i in range(len(main_list) - n + 1):
        if main_list[i:i+n] == sub_list:
            return True
    return False

print(is_sublist([1, 2, 3, 4, 5], [3, 4]))
```

True

42)Write a Python program to split a list into different variables.

Ans:

```
] : lst = [10, 20, 30]
    a, b, c = lst
    print(a, b, c)
```

10 20 30

43)What is tuple? Difference between list and tuple.

Ans:

Tuple:

A tuple is an ordered collection of elements that is **immutable** (cannot be changed).

List	Tuple
Mutable	Immutable
Uses []	Uses ()
Slower	Faster
More memory	Less memory

44)Write a Python program to create a tuple with different data types.

Ans:

```
t = (10, "Python", 3.14, True)
print(t)
```

```
(10, 'Python', 3.14, True)
```

```
|
```

45)Write a Python program to unzip a list of tuples into individual lists.

Ans:

```
data = [(1, 'a'), (2, 'b'), (3, 'c')]
nums, chars = zip(*data)
print(list(nums))
print(list(chars))
```

```
[1, 2, 3]
```

```
['a', 'b', 'c']
```

```
|
```

46)Write a Python program to convert a list of tuples into a dictionary.

Ans:

```
lst = [("a", 1), ("b", 2)]  
d = dict(lst)  
print(d)
```

```
{'a': 1, 'b': 2}
```

|

47)How will you create a dictionary using tuples in python?

Ans:

```
keys = ('a', 'b', 'c')  
values = (1, 2, 3)  
d = dict(zip(keys, values))  
print(d)
```

```
{'a': 1, 'b': 2, 'c': 3}
```

|

48)Write a Python script to sort (ascending and descending) a dictionary by value.

Ans:

```
d = {'a': 3, 'b': 1, 'c': 2}

asc = dict(sorted(d.items(), key=lambda x: x[1]))
desc = dict(sorted(d.items(), key=lambda x: x[1], reverse=True))

print(asc)
print(desc)
```

```
{'b': 1, 'c': 2, 'a': 3}
{'a': 3, 'c': 2, 'b': 1}
```

49)Write a Python script to concatenate following dictionaries to create a new one.

Ans:

```
d1 = {'a': 1}
d2 = {'b': 2}
d3 = {'c': 3}

new_dict = {**d1, **d2, **d3}
print(new_dict)
```

```
{'a': 1, 'b': 2, 'c': 3}
```

50)Write a Python script to check if a given key already exists in a dictionary.

Ans:

```
d = {'a': 1, 'b': 2}
print('a' in d)
```

True

51)How Do You Traverse Through a Dictionary Object in Python?

Ans:

```
d = {'a': 1, 'b': 2}
for key, value in d.items():
    print(key, value)
```

```
a 1
b 2
```

|

52)How Do You Check the Presence of a Key in A Dictionary?

Ans:

```
if 'b' in d:
    print("Key exists")
```

```
Key exists
```

|

53)Write a Python script to print a dictionary where the keys are numbers between 1 and 15.

Ans:

```
d = {i: i*i for i in range(1, 16)}
print(d)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64, 9: 81, 10: 100, 11: 121, 12: 144, 13: 169, 14: 196, 15: 225}
```

|

📄 ⬆

54)Write a Python program to check multiple keys exists in a dictionary

Ans:

```
d = {'a': 1, 'b': 2, 'c': 3}
keys = ['a', 'b']

print(all(k in d for k in keys))
```

True

55)Write a Python script to merge two Python dictionaries

Ans:

```
d1 = {'a': 1}
d2 = {'b': 2}

d1.update(d2)
print(d1)
```

{'a': 1, 'b': 2}

56)Write a Python program to map two lists into a dictionary Sample output: Counter ({'a': 400, 'b': 400,'d': 400, 'c': 300}).

Ans:

```
from collections import Counter

keys = ['a', 'b', 'c', 'd']
values = [400, 400, 300, 400]

result = Counter(dict(zip(keys, values)))
print(result)
```

Counter({'a': 400, 'b': 400, 'd': 400, 'c': 300})

57)Write a Python program to find the highest 3 values in a dictionary

Ans:

```
d = {'a': 10, 'b': 30, 'c': 20, 'd': 40}
top3 = sorted(d.values(), reverse=True)[:3]
print(top3)
```

[40, 30, 20]

58)Write a Python program to combine values in python list of dictionaries. Sample data: [{'item': 'item1', 'amount': 400}, {'item': 'item2', 'amount': 300}, o {'item': 'item1', 'amount': 750}] Expected Output: • Counter ({'item1': 1150, 'item2': 300})

Ans:

```
from collections import Counter

data = [
    {'item': 'item1', 'amount': 400},
    {'item': 'item2', 'amount': 300},
    {'item': 'item1', 'amount': 750}
]

result = Counter()
for d in data:
    result[d['item']] += d['amount']

print(result)
```

Counter({'item1': 1150, 'item2': 300})

59)Write a Python program to create a dictionary from a string. Note: Track the count of the letters from the string.

Ans:

```
from collections import Counter
```

```
s = "w3resource"
```

```
result = dict(Counter(s))
```

```
print(result)
```

```
{'w': 1, '3': 1, 'r': 2, 'e': 2, 's': 1, 'o': 1, 'u': 1, 'c': 1}
```

60)Sample string: 'w3resource' Expected output: • {'3': 1,'s': 1, 'r': 2, 'u': 1, 'w': 1, 'c': 1, 'e': 2, 'o': 1}

Ans:

```
{'w': 1, '3': 1, 'r': 2, 'e': 2, 's': 1, 'o': 1, 'u': 1, 'c': 1}
```

61)Write a Python function to calculate the factorial of a number (a nonnegative integer)

Ans:

```
def fact(n):  
    f = 1  
    for i in range(1, n+1):  
        f = f * i  
    return f  
  
print(fact(5))
```

```
120
```

62)Write a Python function to check whether a number is in a given range

Ans:


```
def check(n):  
    if n >= 1 and n <= 10:  
        return "Yes"  
    else:  
        return "No"  
  
print(check(5))
```

Yes

63)Write a Python function to check whether a number is perfect or not.

Ans:

```
: def perfect(n):  
    s = 0  
    for i in range(1, n):  
        if n % i == 0:  
            s += i  
    if s == n:  
        print("Perfect")  
    else:  
        print("Not Perfect")  
  
perfect(6)
```

Perfect

64)Write a Python function that checks whether a passed string is palindrome or not

Ans:

```
def palindrome(s):  
    if s == s[::-1]:  
        print("Palindrome")  
    else:  
        print("Not Palindrome")  
  
palindrome("madam")
```

Palindrome

65)How Many Basic Types of Functions Are Available in Python?

Ans:

Two types:

1. Built-in functions
2. User-defined functions

66)How can you pick a random item from a list or tuple?

Ans:

```
import random  
a = [1, 2, 3, 4]  
print(random.choice(a))
```

3

67)How can you pick a random item from a range?

Ans:

```
import random
print(random.randrange(1, 10))
```

1

|

68)How can you get a random number in python

Ans:

```
import random
print(random.randint(1, 10))
```

9

69)How will you set the starting value in generating random numbers?

Ans:

```
: import random
random.seed(1)
print(random.randint(1, 10))
```

3

70) How will you randomize the items of a list in place?

Ans:

```
import random
a = [1, 2, 3]
random.shuffle(a)
print(a)
```

[2, 1, 3]

71)What is File function in python? What are keywords to create and write file.

Ans:

- **File function: open()**
- **Modes:**
 - **r → read**
 - **w → write**
 - **a → append**

72)Write a Python program to read an entire text file.

Ans:

```
|: file = open("test.txt", "w")
file.write("Hello Python\nWelcome to File Handling")
file.close()
```

```
|: file = open("test.txt", "r")
print(file.read())
file.close()
```

Hello Python
Welcome to File Handling

73)Write a Python program to append text to a file and display the text.

Ans:

```
f = open("a.txt", "a")
f.write("Hello")
f.close()
```

74)Write a Python program to read first n lines of a file.

Ans:

```
f = open("a.txt", "r")
print(f.readline())
print(f.readline())
f.close()
```

Hello

75)Write a Python program to read last n lines of a file

Ans:

```
f = open("a.txt", "r")
l = f.readlines()
print(l[-2:])
f.close()
```

['Hello']

76)Write a Python program to read a file line by line and store it into a list

Ans:

```
: f = open("a.txt", "r")
l = f.readlines()
f.close()
print(l)
```

['Hello']

77)Write a Python program to read a file line by line store it into a variable.

Ans:

```
f = open("a.txt", "r")
text = f.read()
f.close()
print(text)
```

Hello

78)Write a python program to find the longest words.

Ans:

```
f = open("a.txt", "r")
w = f.read().split()
f.close()
print(max(w, key=len))
```

Hello

79)Write a Python program to count the number of lines in a text file.

Ans:

```
f = open("a.txt", "r")
print(len(f.readlines()))
f.close()
```

1

80)Write a Python program to count the frequency of words in a file.

Ans:

```
f = open("a.txt", "r")
w = f.read().split()
f.close()

for i in w:
    print(i, "=", w.count(i))
```

Hello = 1

81)Write a Python program to write a list to a file.

Ans:

```
lst = [10, 20, 30]

f = open("a.txt", "w")
for i in lst:
    f.write(str(i) + "\n")
f.close()
```

```
f = open("a.txt", "r")
print(f.read())
f.close()
```

10
20
30

82)Write a Python program to copy the contents of a file to another file.

Ans:

```
: f1 = open("a.txt", "r")
  f2 = open("b.txt", "w")

  f2.write(f1.read())

  f1.close()
  f2.close()
```

```
: f = open("a.txt", "w")
  f.write("Hello\nPython File Copy Example")
  f.close()
```

```
: f = open("b.txt", "r")
  print(f.read())
  f.close()
```

```
Hello
Python File Copy Example
```

83)Explain Exception handling? What is an Error in Python?

Ans:

1. Exception handling is used to handle runtime errors.
2. It prevents program from crashing.
3. Error is a problem that stops normal execution

84)How many except statements can a try-except block have? Name Some built-in exception classes:

Ans:

- A try block can have multiple except blocks

Some built-in exceptions:

- ZeroDivisionError
- ValueError
- TypeError

- **FileNotFoundError**
- **IndexError**

85)When will the else part of try-except-else be executed?

Ans:

```
try:
    print(10/2)
except:
    print("Error")
else:
    print("No Error")
```

5.0

No Error

86)Can one block of except statements handle multiple exception?

Ans:

```
try:
    a = int("abc")
except (ValueError, TypeError):
    print("Error handled")
```

Error handled

87)When is the finally block executed?

Ans:

```
: try:
    print(10/0)
except:
    print("Error")
finally:
    print("Always runs")
```

Error
Always runs

88)What happens when „1“== 1 is executed?

Ans:

- Result is False
- Because one is string and other is integer

89)How Do You Handle Exceptions with Try/Except/Finally in Python? Explain with coding snippets.

Ans:

Always runs

```
try:
    a = int(input("Enter number: "))
    print(10/a)
except:
    print("Error occurred")
finally:
    print("Program Ended")
```

Enter number: 33
0.30303030303030304
Program Ended

90)Write python program that user to enter only odd numbers, else will raise an exception.

Ans:

```
try:
    n = int(input("Enter odd number: "))
    if n % 2 == 0:
        raise ValueError("Even number not allowed")
    print("You entered:", n)
except ValueError as e:
    print(e)
```

Enter odd number: 7

You entered: 7

Module 8

DA- Working with NumPy (python)

Assignment

1. How to Creating a 3x3 Identity Matrix with Float Data Type?

Ans:

```
import numpy as np
```

```
np.identity(3, dtype=float)
```

```
array([[1., 0., 0.],  
       [0., 1., 0.],  
       [0., 0., 1.]])
```

2. Create a 1D Array with Random Values between 0 and 1.

Ans:

```
np.random.rand(5)
```

```
array([0.7150833 , 0.91616488, 0.90331418, 0.70312506, 0.531469  ])
```

3. Create a 2D Array with Random Integer Values.

Ans:

```
|: np.random.randint(1, 10, size=(3,3))
```

```
|: array([[5, 7, 8],  
         [7, 3, 7],  
         [7, 7, 6]])
```

```
|: 
```

4. Creating an Array Using a Custom Function.

Ans:

```
] import numpy as np

def square(i, j):
    return i * j

np.fromfunction(square, (3,3), dtype=int)

:] array([[0, 0, 0],
          [0, 1, 2],
          [0, 2, 4]])
```

1.

5. Reshaping a 1D Array into a 2D Array

Ans:

```
: a = np.arange(6)
  a.reshape(2,3)

:] array([[0, 1, 2],
          [3, 4, 5]])
```

:

6.How to Creating a 3x3 Array of Ones?

Ans:

```
np.ones((3,3))

array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

7. How to get the common items between two python's NumPy? Input: a = np. array ([1,2,3,2,3,4,3,4,5,6]) b = np. array ([7,2,10,2,7,4,9,4,9,8]) Expected Output: array ([2, 4]) 8. From array a remove all items

Ans:

```
a = np.array([1,2,3,2,3,4,3,4,5,6])
b = np.array([7,2,10,2,7,4,9,4,9,8])

np.intersect1d(a, b)
```

```
array([2, 4])
```

8. From array a remove all items present in array b Input: a = np. array ([1,2,3,4,5]) b = np. array ([5,6,7,8,9]) Expected Output: array ([1,2,3,4])

Ans:

```
a = np.array([1,2,3,4,5])
b = np.array([5,6,7,8,9])

np.setdiff1d(a, b)
```

```
array([1, 2, 3, 4])
```

9. Limit the number of items printed in python NumPy array a to a maximum of 6 elements. a = np. arrange (15) Expected Output: array ([0, 1, 2, ..., 12, 13, 14])

Ans:

```
np.set_printoptions(threshold=6)
a = np.arange(15)
a
```

```
array([ 0,  1,  2, ..., 12, 13, 14])
```

10. Drop all nan values from a 1D NumPy array Input: np. array ([1,2,3, np.nan,5,6,7, np.nan]) Desired Output: array ([1., 2., 3., 5., 6., 7.])

Ans:

```
a = np.array([1,2,3,np.nan,5,6,7,np.nan])
a[~np.isnan(a)]
```

```
array([1., 2., 3., 5., 6., 7.])
```

11. Question: Create a 1D NumPy array of the first 20 natural numbers and a 2D NumPy array of shape (4, 5) with values ranging from 1 to 20.

Ans:

```
a = np.arange(1, 21)
print("1D Array:")
print(a)
b = a.reshape(4, 5)
print("\n2D Array (4x5):")
print(b)
```

```
1D Array:
[ 1  2  3 ... 18 19 20]
```

```
2D Array (4x5):
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]
 [11 12 13 14 15]
 [16 17 18 19 20]]
```

12. Question: Given a 3D NumPy array of shape (2, 3, 4), find its shape, size, number of dimensions, and data type. Change its data type to float64 and verify the change.

Ans:

```
a = np.random.randint(1,10,(2,3,4))

a.shape
a.size
a.ndim
a.dtype

a = a.astype(np.float64)
a.dtype
```

```
dtype('float64')
```


13. Question: Reshape a 1D array of 12 elements into a 3x4 2D array and then flatten it back into a 1D array using `ravel()`. Verify that the flattened array matches the original.

Ans:

```
a = np.arange(12)
b = a.reshape(3,4)
c = b.ravel()

np.array_equal(a, c)
```

True

14. Question: Given two arrays, `a = np. array ([1, 2, 3])` and `b = np. array ([4, 5, 6])`, perform element-wise addition, subtraction, multiplication, and division. Explain the behavior when dividing by zero.

Ans:

```
a = np.array([1,2,3])
b = np.array([4,5,6])
```

```
a+b
a-b
a*b
a/b
```

```
array([0.25, 0.4 , 0.5 ])
```

15. Question: Create a 2D array of shape (3, 1) and a 1D array of length 3. Perform element-wise addition using broadcasting. Explain how broadcasting rules apply in this scenario.

Ans:

```
a = np.array([[1],[2],[3]])
b = np.array([10,20,30])

a + b
```

```
array([[11, 21, 31],
       [12, 22, 32],
       [13, 23, 33]])
```

```
|
```

16. Question: Generate a random 2D array of integers between 0 and 10.

Ans:

```
np.random.randint(0,11,(3,3))
|
```

```
array([[0, 2, 2],
       [5, 4, 5],
       [1, 0, 8]])
```

17. Question: Given a 4x4 array of random integers, use indexing and slicing to extract:

- o The entire second row
- o The last column
- o The subarray consisting of the first two rows and first two columns

Ans:

```
a = np.random.randint(1,20,(4,4))

a[1]
a[:, -1]
a[:2, :2]
```

```
array([[17,  7],
       [17,  4]])
```

18. Question: Describe a practical example where NumPy can be used in EDA, AI, ML, and DL, and implement a NumPy solution for a simple task in each area.

Ans:

```
np.mean([10,20,30])

np.array([1,2]) + np.array([3,4])

x = np.array([10,20,30])
(x - x.mean()) / x.std()

np.dot([[1,2],[3,4]], [[5,6],[7,8]])

array([[19, 22],
       [43, 50]])
```

19. Question: Create a random 4x4 matrix and use NumPy to compute its eigenvalues and eigenvectors. Verify the eigenvalues by reconstructing the matrix.

Ans:

```
a = np.random.rand(4,4)
vals, vecs = np.linalg.eig(a)
vals

array([ 1.48594274+0.j, -0.04454308+0.29267567j,
       -0.04454308-0.29267567j, -0.05564736+0.j])
```

20. Question: Create a 1D array of 27 elements and reshape it into a 3x3x3 3D array. Flatten it back into a 1D array and compare the flattened array with the original.

Ans:

```
a = np.arange(27)
b = a.reshape(3,3,3)
c = b.flatten()

np.array_equal(a, c)
```

True

21. Question: Perform matrix multiplication of two 2D arrays using np.dot () and @. Compare the results and performance of both methods using a large dataset.

Ans:

```
a = np.random.rand(100,100)
b = np.random.rand(100,100)

np.dot(a,b)
a @ b
```

```
array([[26.53663218, 24.94562307, 27.32966289, ..., 27.12366994,
        25.48467803, 25.72954308],
       [23.51081121, 21.03058601, 23.91878879, ..., 22.57248668,
        23.1740433 , 23.8527814 ],
       [26.48823623, 23.42576621, 27.54438711, ..., 24.00179999,
        26.71945768, 24.32618422],
       ...,
       [24.98438165, 21.56852469, 26.21366485, ..., 24.1024843 ,
        22.82954741, 23.61017201],
       [27.89879075, 24.95052806, 27.94507002, ..., 26.76939611,
        26.87058776, 26.86817547],
       [26.77880646, 23.86144573, 27.60841161, ..., 24.03680092,
        25.87755367, 25.69394654]])
```

22. Question: Create a 3D array of shape (2, 1, 4) and a 2D array of shape (4, 1). Perform an element-wise operation using broadcasting and explain the result. Use np. new axis to achieve the same result without broadcasting.

Ans:

```
a = np.random.rand(2,1,4)
b = np.random.rand(4,1)

a + b.T
```

```
array([[ [1.18829918, 0.92748424, 0.76353332, 1.37223782]],
       [ [1.26511694, 1.4018343 , 0.42326569, 1.18683277]]])
```

23. Question: Generate a 2D array of random floats between 0 and 1. Use conditional operators to create a Boolean mask for values less than 0.5. Replace these values with their squares and leave the rest unchanged.

Ans:

```
a = np.random.rand(4,4)
mask = a < 0.5
a[mask] = a[mask] ** 2
a
```

```
array([[0.12296784, 0.01138046, 0.23738085, 0.01074003],
       [0.0905586 , 0.2179338 , 0.68072034, 0.93880724],
       [0.12391327, 0.79570735, 0.54275431, 0.16593783],
       [0.04156943, 0.57307177, 0.15185973, 0.94889072]])
```

24. Question: Given a 5x5 array of sequential integers, use slicing to: o Extract the diagonal elements o Replace the elements of the middle row with zeros o Flip the array vertically and horizontally

Ans:

```
a = np.random.randint(1,10,(2,3,4,5))
a[:,2,:,:].mean(axis=2)
```

```
array([[[5.25, 5.25, 5.75, 4.5 , 3.75],
       [5. , 4. , 5.75, 5.25, 5. ]],
       [[6. , 4. , 5.75, 5.25, 6. ],
       [5.75, 6.5 , 7.25, 2.75, 5. ]]])
```

25. Question: Create a 4D array of shape (2, 3, 4, 5) with random integers. Use advanced slicing to extract a subarray and compute the mean along a specified axis.

Ans:

```
a = np.random.randint(1,10,(2,3,4,5))
a[:,2,:,:].mean(axis=2)

array([[2.25, 5.5 , 5.75, 3.5 , 6.25],
       [6. , 5. , 5.25, 7. , 7.75]],

      [[2.5 , 5. , 2.75, 6.25, 5. ],
       [7. , 5.25, 3.25, 3.75, 5.25]])
```

26. Question: Given an array of shape (10, 20), reshape it to (20, 10) and (5, 40). Discuss the impact on the array's shape, size, and dimensionality.

Ans:

```
: a = np.arange(200).reshape(10,20)
a.reshape(20,10)
a.reshape(5,40)

: array([[ 0,  1,  2, ..., 37, 38, 39],
       [40, 41, 42, ..., 77, 78, 79],
       [80, 81, 82, ..., 117, 118, 119],
       [120, 121, 122, ..., 157, 158, 159],
       [160, 161, 162, ..., 197, 198, 199]])
```

27. Question: Generate a large 2D array and demonstrate the use of np. reshape () and unravel () to manipulate its shape for various linear algebra operations.

Ans:

```
|: a = np.arange(100).reshape(10,10)
   np.unravel_index(45, a.shape)
```

```
|: (4, 5)
```

```
|: 
```

28. Question: Given a 6x6 matrix, use advanced indexing and slicing to extract the upper triangular part of the matrix and set the lower triangular part to zero. Verify the result.

Ans:

```
a = np.random.randint(1,10,(6,6))
a[np.tril_indices(6, -1)] = 0
a
```

```
array([[3, 9, 2, 6, 1, 8],
       [0, 8, 5, 7, 3, 4],
       [0, 0, 7, 7, 6, 2],
       [0, 0, 0, 9, 7, 1],
       [0, 0, 0, 0, 5, 4],
       [0, 0, 0, 0, 0, 1]])
```

Module 9)
DA- Working with Pandas (python)
Assignment

1) Create a series of three different colors

2) View the series of different colors

Ans:

```
import pandas as pd
```

```
colors = pd.Series(["Red", "Blue", "Green"])
```

```
print(colors)
```

```
0    Red
1    Blue
2    Green
dtype: object
```

3) Create a series of three different car types and view it

Ans:

```
cars = pd.Series(["BMW", "Audi", "Toyota"])
print(cars)
```

```
0    BMW
1    Audi
2    Toyota
dtype: object
```

4) Combine the Series of cars and colors into a Data Frame

Ans:


```
car_sales = pd.DataFrame({  
    "Car": cars,  
    "Color": colors  
})  
  
print(car_sales)
```

	Car	Color
0	BMW	Red
1	Audi	Blue
2	Toyota	Green

5) Find the different datatypes of the car data Data Frame

Ans:

```
car_sales.dtypes
```

```
Car      object  
Color    object  
dtype: object
```

6) Describe your current car sales Data Frame using describe ()

Ans:

```
car_sales.describe(include="all")
```

	Car	Color
count	3	3
unique	3	3
top	BMW	Red
freq	1	1

7) Get information about your Data Frame using info ()

Ans:

```
car_sales.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  -
0    Car      3 non-null      object
1   Color    3 non-null      object
dtypes: object(2)
memory usage: 180.0+ bytes
```

8) Create a Series of different numbers and find the mean of them

Ans:

```
nums = pd.Series([10, 20, 30, 40])
print(nums.mean())
```

25.0

9) Create a Series of different numbers and find the sum of them

Ans:

```
|: print(nums.sum())
```

100

10) List out all the column names of the car sales Data Frame

Ans:

```
: car_sales = pd.DataFrame({
    "Make": ["Toyota", "BMW", "Audi", "Toyota", "BMW", "Audi", "Toyota"],
    "Color": ["Red", "Blue", "Black", "White", "Red", "Blue", "Black"],
    "Doors": [4, 4, 2, 4, 2, 4, 4],
    "Price": [10000, 20000, 25000, 12000, 22000, 26000, 13000]
})

print(car_sales)
```

	Make	Color	Doors	Price
0	Toyota	Red	4	10000
1	BMW	Blue	4	20000
2	Audi	Black	2	25000
3	Toyota	White	4	12000
4	BMW	Red	2	22000
5	Audi	Blue	4	26000
6	Toyota	Black	4	13000

```
: car_sales.columns
```

```
: Index(['Make', 'Color', 'Doors', 'Price'], dtype='object')
```

11) Find the length of the car sales Data Frame

Ans:

```
len(car_sales)
```

7

12) Show the first 5 rows of the car sales Data Frame

Ans:

```
|: car_sales.head()
```

```
|:      Make  Color  Doors  Price
0  Toyota   Red     4  10000
1   BMW    Blue     4  20000
2   Audi   Black     2  25000
3  Toyota  White     4  12000
4   BMW    Red      2  22000
```

13) Show the first 7 rows of the car sales Data Frame

Ans:

```
car_sales.head(7)
```

```
      Make  Color  Doors  Price
0  Toyota   Red     4  10000
1   BMW    Blue     4  20000
2   Audi   Black     2  25000
3  Toyota  White     4  12000
4   BMW    Red      2  22000
5   Audi   Blue     4  26000
6  Toyota  Black     4  13000
```

14) Show the bottom 5 rows of the car sales Data Frame

```
car_sales.tail()
```

	Make	Color	Doors	Price
2	Audi	Black	2	25000
3	Toyota	White	4	12000
4	BMW	Red	2	22000
5	Audi	Blue	4	26000
6	Toyota	Black	4	13000

15) Use. loc to select the row at index 3 of the car sales Data Frame

Ans:

```
car_sales.loc[3]
```

```
Make      Toyota
Color      White
Doors         4
Price     12000
Name: 3, dtype: object
```

16) Use. iloc to select the row at position 3 of the car sales Data Frame

Ans:

```
car_sales.iloc[3]
```

```
Make      Toyota
Color      White
Doors         4
Price     12000
Name: 3, dtype: object
```

17) Create a crosstab of the Make and Doors columns.

Ans:

```
pd.crosstab(car_sales["Make"], car_sales["Doors"])
```

Doors 2 4

Make

Audi 1 1

BMW 1 1

Toyota 0 3

Module

10) DA- Visualization with Matplotlib and Seaborn

Assignment Dataset contains information about some police deaths in US from 1984 to 2016. Create the following visualizations using this dataset.

1) Bar chart showing the total deaths per year for 1984-2016

Ans:

```
7]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
9]: df = pd.read_csv("C:/Users/sonyd/Downloads/U.S.Death.csv")
print(df.head())
```

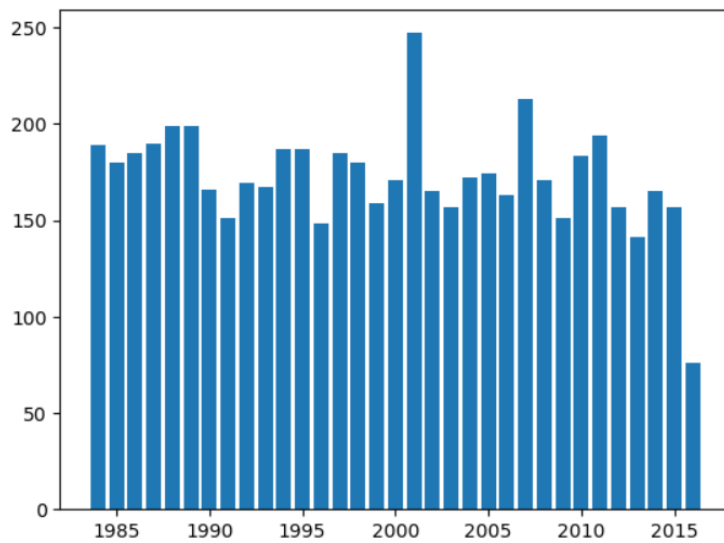
```

           person \
0   Constable Darius Quimby
1  Sheriff Cornelius Hogeboom
2  Deputy Sheriff Isaac Smith
3      Marshal Robert Forsyth
4   Sheriff Robert Maxwell

           dept \
0   Albany County Constable's Office, NY
1   Columbia County Sheriff's Office, NY
2   Westchester County Sheriff's Department, NY
3   United States Department of Justice - United S...
4   Greenville County Sheriff's Office, SC

           eow           cause cause_short \
0   EOW: Monday, January 3, 1791  Cause of Death: Gunfire  Gunfire
1   EOW: Saturday, October 22, 1791  Cause of Death: Gunfire  Gunfire
2   EOW: Thursday, May 17, 1792  Cause of Death: Gunfire  Gunfire
3   EOW: Saturday, January 11, 1794  Cause of Death: Gunfire  Gunfire
4   EOW: Sunday, November 12, 1797  Cause of Death: Gunfire  Gunfire
```

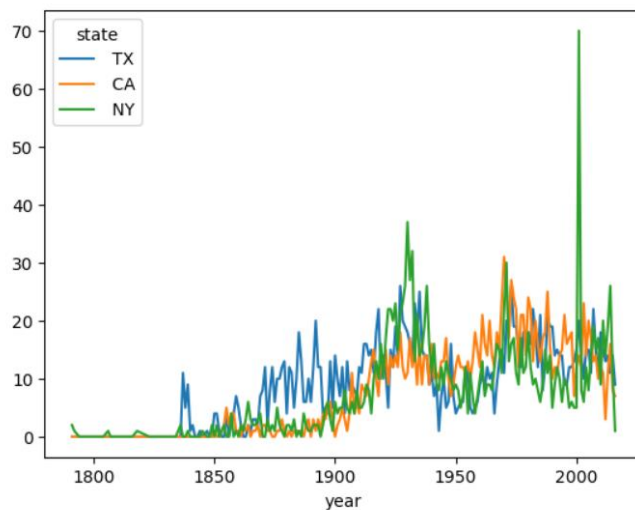
```
y = df[(df['year']>=1984) & (df['year']<=2016)][ 'year'].value_counts().sort_index()
plt.bar(y.index, y.values)
plt.show()
```



2) Line chart comparing the yearly deaths in different states

Ans:

```
top = df['state'].value_counts().head(3).index
df.groupby(['year', 'state']).size().unstack(fill_value=0)[top].plot()
plt.show()
```



3) A heat map overlaid on the map of the United States

Ans:


```
df['state'].unique()
```

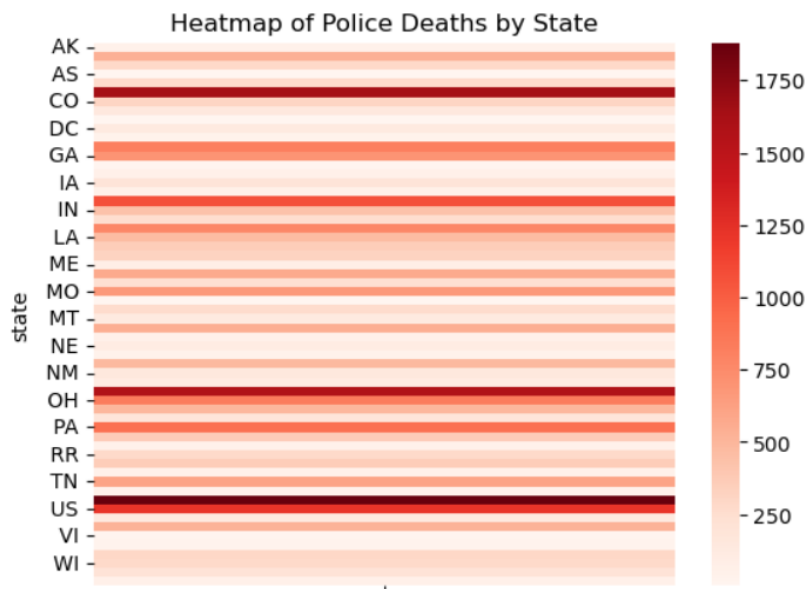
```
array([' NY', ' US', ' SC', ..., ' GU', ' AS', ' MP'], dtype=object)
```

4) A Choropleth comparing the number of police deaths in different states

ans:

```
: state_counts = df.groupby('state').size().to_frame('deaths')
```

```
plt.figure()
sns.heatmap(state_counts, cmap="Reds", cbar=True)
plt.title("Heatmap of Police Deaths by State")
plt.show()
```



5) A word cloud of the different causes of death (remove the string "Cause of Death:" for best results)

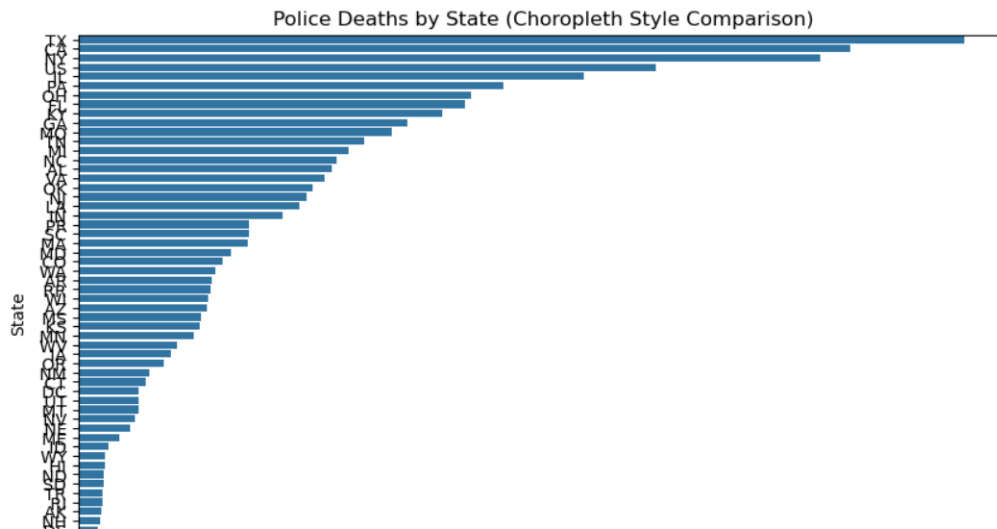
Ans:

```

]: state_counts = df.groupby('state').size().sort_values(ascending=False)

plt.figure(figsize=(10,6))
sns.barplot(x=state_counts.values, y=state_counts.index)
plt.title("Police Deaths by State (Choropleth Style Comparison)")
plt.xlabel("Total Deaths")
plt.ylabel("State")
plt.show()

```



6) A marker cluster showing the shootings in the state of California (show person's name on hover)

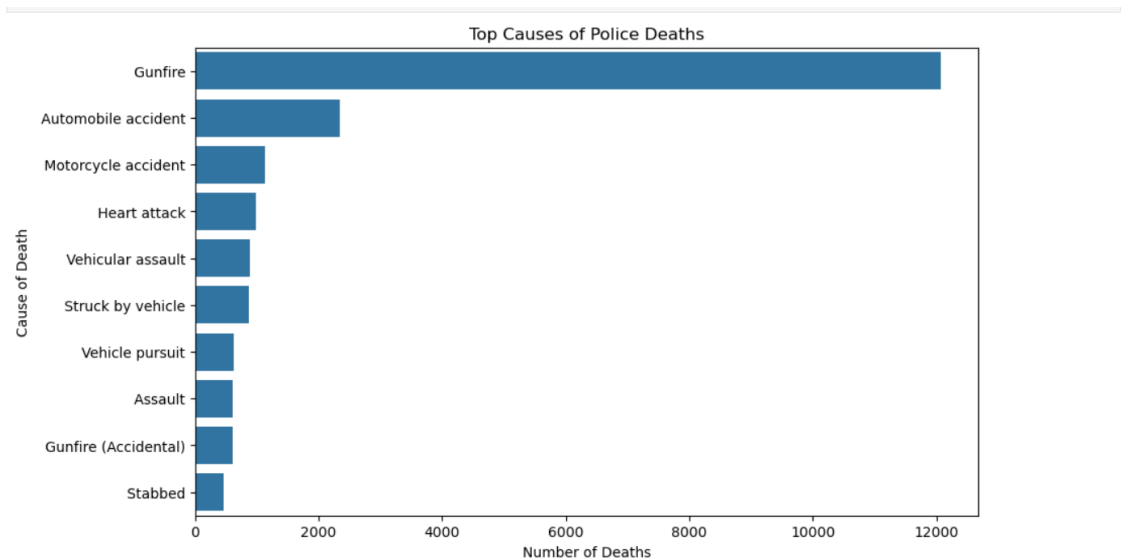
Ans:

```

cause_counts = (
    df['cause']
    .str.replace("Cause of Death:", "", regex=False)
    .value_counts()
    .head(10)
)

plt.figure(figsize=(10,6))
sns.barplot(x=cause_counts.values, y=cause_counts.index)
plt.xlabel("Number of Deaths")
plt.ylabel("Cause of Death")
plt.title("Top Causes of Police Deaths")
plt.show()

```



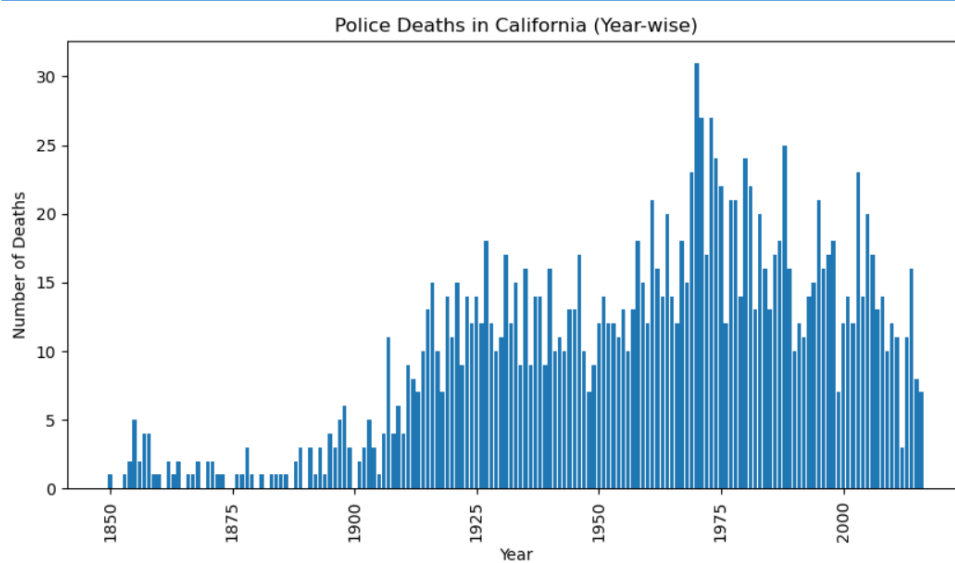
7) Bar chart comparing the no. of deaths due to different causes, animated by year

Ans:

```
ca_data = df[df['state'].str.contains('CA', case=False, na=False)]

yearly_ca = ca_data.groupby('year').size()

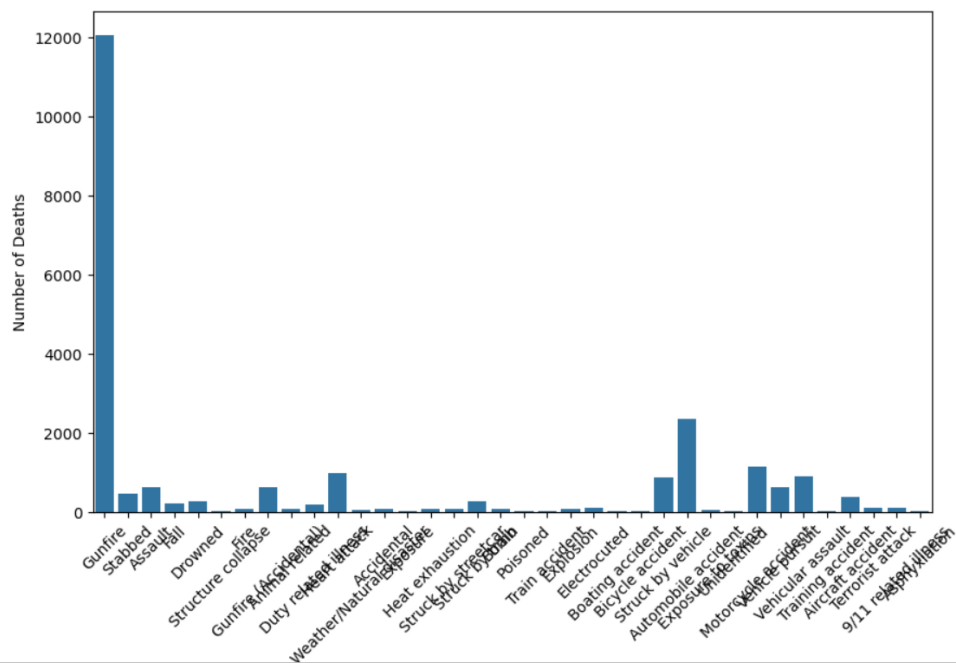
plt.figure(figsize=(10,5))
plt.bar(yearly_ca.index, yearly_ca.values)
plt.xlabel("Year")
plt.ylabel("Number of Deaths")
plt.title("Police Deaths in California (Year-wise)")
plt.xticks(rotation=90)
plt.show()
```



8) A heatmap of total deaths per state per year i.e., showing "state" on one axis and "year" on the other axis.

Ans:

```
] : plt.figure(figsize=(10,6))
sns.countplot(data=df, x='cause_short')
plt.xticks(rotation=45)
plt.xlabel("Cause of Death")
plt.ylabel("Number of Deaths")
plt.title("Deaths Due to Different Causes")
plt.show()
```

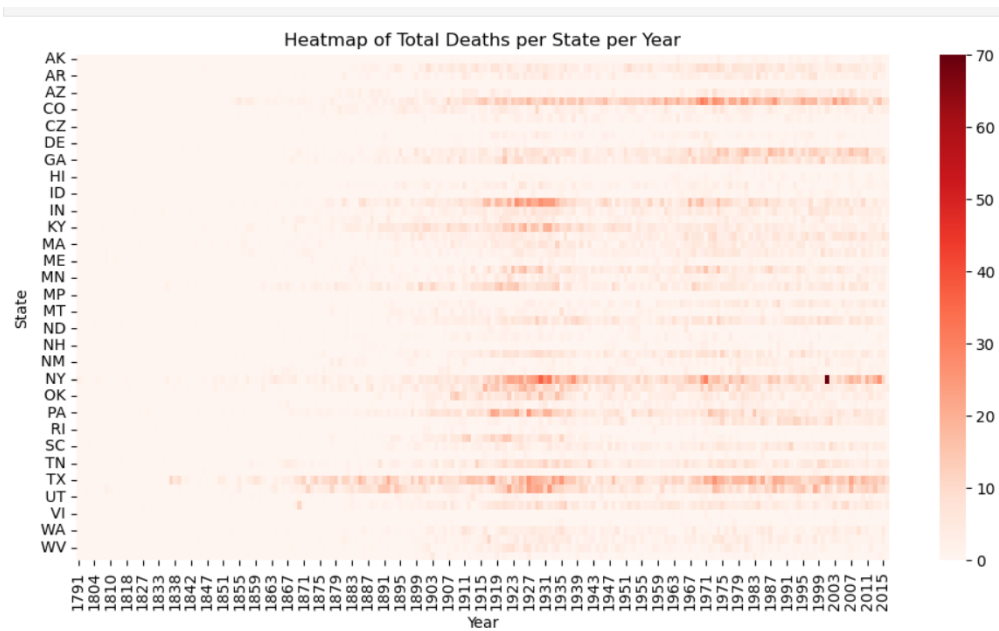


9) A tree map with three levels: state, city (description), cause

Ans:

```
heat = df.pivot_table(index='state', columns='year', aggfunc='size', fill_value=0)

plt.figure(figsize=(12,6))
sns.heatmap(heat, cmap="Reds")
plt.xlabel("Year")
plt.ylabel("State")
plt.title("Heatmap of Total Deaths per State per Year")
plt.show()
```



10) A rug plot showing a timeline of canine deaths.

Ans:

```
import matplotlib.dates as mdates
canine_df['date'] = pd.to_datetime(canine_df['date'])
deaths_per_day = canine_df.groupby('date').size()

plt.figure(figsize=(12, 5))
deaths_per_day.plot(kind='line', color='orange')
plt.title("Daily Canine Death Counts")
plt.xlabel("Date")
plt.ylabel("Number of Deaths")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

